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Strings

A string is defined as a sequence of characters where each character may be a letter, number or a symbol. Each element occupies one byte of memory. Every string is terminated by a null ('\0', ASCII code 0) character which must be appended at the end of the string. In C++, there is no basic data type to represent a string. Instead, it implements a string as an one-dimensional character array. When declaring a character array, it also has to hold a null character at the end, and so, the size of the character array should be one character longer than the length of the string.

Character Array (String) creation

To create any kind of array, the size (length) of the array must be known in advance, so that the memory locations can be allocated according to the size of the array. Once an array is created, its length is fixed and cannot be changed during run time. This is shown in figure12.2

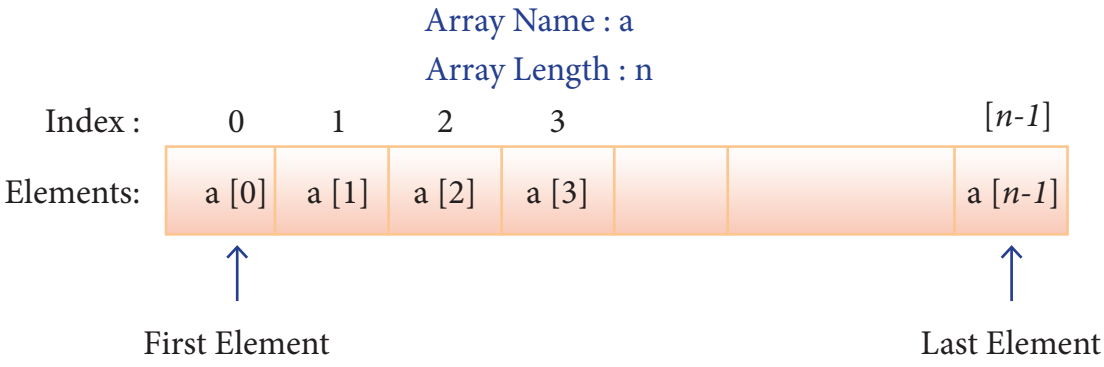

Figure 12.2

Syntax

Array declaration is:

char array_name[size];

In the above declaration, the size of the array must be an unsigned integer value.

For example,

char country[6];

Here, the array reserves 6 bytes of memory for storing a sequence of characters. The length of the string cannot be more than 5 characters and one location is reserved for the null character at the end.